

Suwisa Aitha

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Summary

M.S. in Electrical Engineering from George Mason University with a concentration in Power Systems and Smart Grids. Experienced in power system analysis, power quality studies, renewable energy integration, and engineering simulation using MATLAB/Simulink and PowerWorld. Strong analytical and problem-solving skills developed through graduate projects and industry experience. Seeking entry-level Electrical Engineer opportunity. Open to relocation.

EDUCATION

George Mason University

M.S. in Electrical Engineering, GPA: 3.22/4.00

Fairfax, VA

Jan 2025 – Graduated May 2026

Khon Kaen University

B.S. in Electrical Engineering

Khon Kaen, Thailand

Aug 2018 – Jul 2022

TECHNICAL SKILLS

Electrical Engineering: Power Flow Analysis, Fault Analysis, Power System Protection, Stability Analysis, Frequency Response, RoCoF Analysis, Harmonic Analysis (THD)

Software: MATLAB/Simulink, SPICE, PowerWorld, AutoCAD, Python, MS Office Suite

Hardware: ESP32, Arduino

PROFESSIONAL EXPERIENCE

Electrical Design and Visualization (Co-op)

Bangkok, Thailand

B.S.E. Electronics Co., Ltd.

Jul 2021 – Nov 2021

- Developed electrical layouts and visualization models using AutoCAD to support electrical system integration
- Compared simulation results with physical measurements and achieved approximately 87% correlation for system validation
- Assisted in preparing technical documentation, electrical configurations, and engineering support tasks
- Supported coordination between design and implementation processes for electrical systems

ENGINEERING PROJECT EXPERIENCE

Grid-Forming Inverter with Virtual Synchronous Machine (VSM) Control

George Mason University

- Developed a grid-forming inverter model with VSM control in MATLAB/Simulink for inverter-based microgrids
- Simulated virtual inertia and frequency support to improve renewable energy integration and grid resilience
- Analyzed inverter synchronization, transient response, and frequency regulation under dynamic conditions

Field Oriented Control (FOC) of Three-Phase Induction Motor – MATLAB/Simulink

George Mason University

- Designed and implemented FOC for a three-phase induction motor using MATLAB/Simulink
- Applied Clarke and Park transformations, PI controllers, and PWM inverter control for speed regulation
- Evaluated motor speed tracking, current response, and dynamic performance under varying conditions

Unified Power Quality Conditioner (UPQC)

George Mason University

- Designed and simulated a UPQC system in MATLAB/Simulink to improve electrical power quality.
- Reduced total harmonic distortion by approximately 53% through active compensation techniques.
- Evaluated system performance during voltage sag and varying load conditions.

Power System Dynamics with Renewable Energy Penetration - Simulation Project

George Mason University

- Developed a Single Machine Infinite Bus (SMIB) model in MATLAB/Simulink for power system dynamic analysis
- Evaluated frequency nadir, rotor angle stability, settling time, and RoCoF during disturbances
- Assessed the impact of renewable energy penetration on power system stability and dynamic performance

CERTIFICATIONS & PROFESSIONAL AFFILIATIONS

- Dominion Energy Fault Analysis Certificate Program – Completed
- IEEE Power & Energy Society (PES) Member

AWARDS & RECOGNITION

- Mason Recreation Fitness Employee of the Year 2025 – 2026